trees were originally invented, they just happened to work very well on this domain, but these containers were meant to solve problems in higher dimensional spaces, rather than in 2-D or 3-D space.

To give you an idea of the vastity of the fields that can leverage these algorithms, we'll briefly go through a few examples on very different domains.

11.4.1 Color reduction

The problem is simple: you have an RGB bitmap image using a certain palette, for example the usual 16 million RGB, where each pixel of the image has 3 channels associated with it, red, green and blue, and each of these colors has associated an intensity value between 0 and 255. For instance, if a pixel is meant to be completely red, it will be associate with the tuple (R=255, G=0, B=0)[12], a darker blue pixel with something like (0, 0, 146), and a yellow one with (255, 255, 0); black is (0, 0, 0) and white, instead (255, 255, 255).

To store such an image, we need 3 bytes per pixel[13]; for an image with a resolution of 1280 x 720 pixels[14], this means 2.7649.800 bytes, more than 2 MBs, and 1920 × 1080 pixels[15] images require almost 6 MBs at 16M colors.

To save space, you could either use a compressed format, which could cause a loss of information or transform the data to a different space[16], or reduce the numbers of colors you use for each pixel. Suppose that you are storing images for a